

Classification and Model Evaluation

Predict categories and measure the different consequences of classification errors.



WHY THIS MATTERS

Start from what you already know

Classification versus regression, linear scores, probability, and test data

- 01 Explain logistic regression in plain language.
- 02 Use probabilities and thresholds in a small worked example.
- 03 Complete the guided lab and verify precision and recall.

CORE CONCEPT

The vocabulary of this topic

Predict categories and measure the different consequences of classification errors.

logistic regression: Logistic regression maps a linear score to a probability for a binary outcome.

probabilities and thresholds: A probability represents uncertainty; a threshold converts that probability into an action or class.

confusion matrix: A confusion matrix counts correct and incorrect predictions for each actual and predicted class.

precision and recall: Precision measures positive prediction reliability; recall measures coverage of actual positives.

FOUR CONNECTED IDEAS

How the concept works

01

**logistic
regression**



02

**probabilities and
thresholds**



03

**confusion
matrix**



04

**precision and
recall**

$P(y=1|x)=1/(1+e^{-(\beta_0+\beta^Tx)})$; a threshold turns probability into a class decision.

CORE CONCEPT

Read every part of the mechanism

$P(y=1|x)=1/(1+e^{-(\beta_0+\beta^T x)})$; a threshold turns probability into a class decision.

probability: A number from 0 to 1.

threshold: The action boundary.

precision: Correct positives among predicted positives.

recall: Found positives among actual positives.

WORKED EXAMPLE

Worked example by hand

From $TP=8$, $FP=2$, $FN=4$, calculate $\text{precision}=0.80$, $\text{recall}\approx 0.67$, and $F1\approx 0.73$.

01

Known values

02

Apply the mechanism

03

Interpret the result

RUN IT AND INSPECT THE RESULT

Map the concept to Python

```
tp, fp, fn = 8, 2, 4
precision = tp / (tp + fp)
recall = tp / (tp + fn)
f1 = 2 * precision * recall / (precision + recall)
print(round(precision, 2), round(recall, 2), round(f1, 2))
```

OUTPUT

0.8 0.67 0.73

Each line corresponds to a concept already explained.

QUALITY GATE

Build the complete mental model



Logistic regression converts a linear score into a probability with the sigmoid function.



Changing the threshold trades false positives against false negatives without retraining.



Accuracy can hide failure on a rare class; precision and recall expose different error costs.



Use MAE/RMSE/ R^2 for regression and confusion-derived metrics for classification.

FOUR CONNECTED IDEAS

Notebook readiness map

01



**logistic
regression**



02

**probabilities and
thresholds**



03

**confusion
matrix**



04

**precision and
recall**

Use the concepts from this lesson to read and complete
<https://github.com/curiously/AI-Bootcamp/blob/master/04.model-evaluation-techniques.ipynb>.

RISK REVIEW

Common beginner mistakes

01 Using accuracy for a rare but important class.

02 Changing the threshold after viewing the final test result.

GUIDED LAB

Evaluate a binary classifier at two thresholds and explain the operational tradeoff.

01

Predict

02

Run

03

Inspect

04

Explain

SESSION SYNTHESIS

Keep the system, not isolated definitions.

01 Explain logistic regression in plain language.

02 Use probabilities and thresholds in a small worked example.

03 Complete the guided lab and verify precision and recall.

NEXT EVIDENCE

An evaluation notebook with a confusion matrix, metric calculations, threshold comparison, and recommendation.